GONIA · γωνία · measured ancient architecture

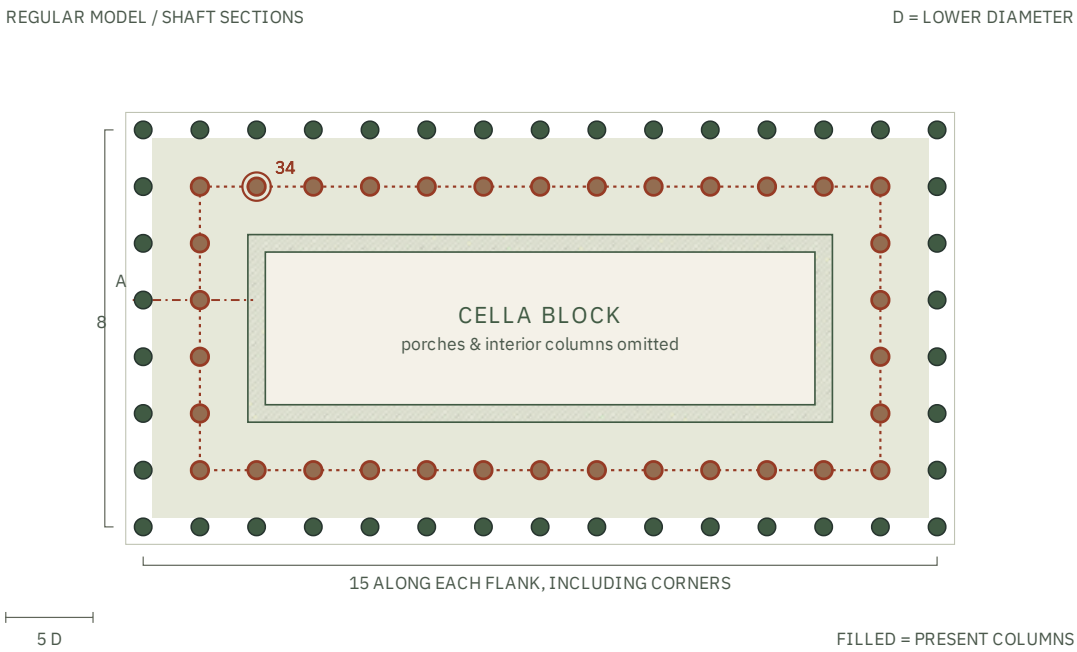
The row that is not there.

Selected arrangement: inner ring omitted. Chosen equal gap 2.25 D; diameter 1 D; axis pitch 3.25 D. The simplified cella block and outer axes stay fixed when the inner ring is omitted.

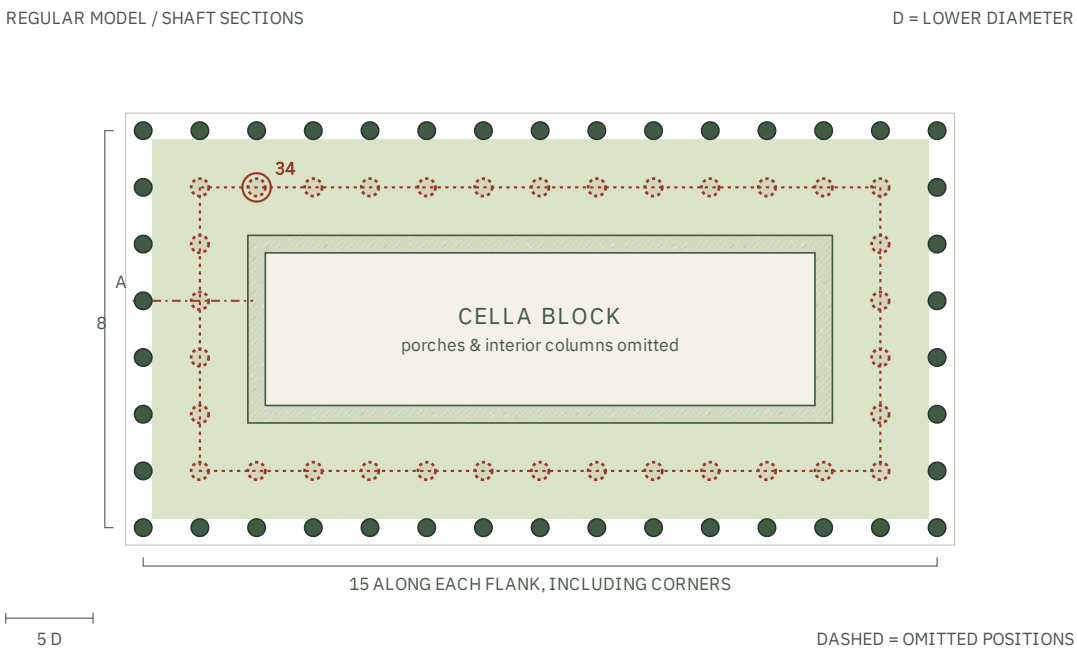
Edition 26785172528696adbca7ded0b578e6143bf0a6dbf16b0aaf2dd094806476fe20. Reopen these settings. Both plans, vocabulary, fonts and all five complete source images are embedded for offline reading.

Compare the two arrangements

Two perimeter rings: 76 model columns



Inner ring omitted: 42 model columns



The selected passage at a column station



2.25 + 1 + 2.25 = 5.5 D. Removing the inner shaft joins the full width. In the double row the lanes communicate through the gaps between columns. The strip is measured between shaft faces, excluding bases.

Distinct model ring positions; corners assigned to ends

Ring	Ends	Flanks without corners	Total
Outer	2 × 8	2 × 13	42
Inner	2 × 6	2 × 11	34

Selected trace: inner ring, 34 positions visited. The plans are regular-grid experiments, not a measured Magnesia reconstruction. Porches, interior columns, bases, steps and roof supports are omitted.

A number the grid cannot settle

Krohn prints XXXIV (34); the apparatus on p. 66 attributes that reading to Philander and records XXXVIII (38) in the textual tradition. The model's regular inner ring has 34 positions. Agreement with a neat grid does not prove that an editorial reading is correct.

Pavlos Lefas (2001) argues for 38 through an interpretation involving the inner building's ends and antae. The model omits its porches. Neither this model nor the photograph establishes that a completed double colonnade was physically removed at Magnesia.

Read the Latin

III.2.6 · Count the ends and the flanks

pseudodipteros autem sic conlocatur, ut in fronte et postico sint columnae octonae, in lateribus cum angularibus quinae denae.

The pseudodipteral temple is arranged so that there are eight columns each at the front and rear, and fifteen on each flank, including the corners.

III.2.6 · Measure between faces

sint autem parietes cellae contra quaternas columnas medianas in fronte et postico. ita duorum intercolumniorum et imae crassitudinis columnae spatium erit ab parietibus circa ad extremos ordines columnarum.

The cella walls are to stand opposite the four middle columns at the front and rear. Thus a space of two intercolumniations and the lower thickness of a column will extend all round from the walls to the outer rows of columns.

III.2.7, excerpt · Read what is doubled

sed circa aedem duplices habet ordines columnarum

But around the temple it has double rows of columns.

III.3.8, excerpt · Remove the inner rows

ex dipteri enim aedis symmetria distulit interiores ordines columnarum XXXIV eaque ratione sumptus operasque compendii fecit.

For from the proportioned arrangement of the dipteral temple he removed the inner rows of thirty-four columns, and in this way saved expense and labour.

III.3.8, excerpt · Keep room for a walk

is in medio ambulationi laxamentum egregie circa cellam fecit de aspectuque nihil inminuit

He provided ample room in the middle for walking around the cella and diminished nothing of its appearance.

Keep the word notes

octonae · eight in each place

A distributive numeral, feminine plural with columnae: eight at the front and eight at the rear. It does not mean eight divided between both ends.

---

quinae denae · fifteen on each side

Two words form the distributive fifteen. The next phrase includes the corner columns, so the long-side count overlaps the end counts.

---

cum angularibus · including the corner columns

Cum takes the ablative. Angularibus stands for the corner columnae. Count each corner once when finding a total, even though it belongs to an end and a flank.

---

duorum intercolumniorum · of two spaces between columns

A genitive quantity attached to spatium, space. The following genitive adds one lower shaft thickness. If a clear gap is g diameters, the passage is 2g + 1, rather than two axis pitches.

---

interiores ordines · the inner rows

The accusative plural is the object of distulit, removed. Columnarum is genitive plural, of columns. XXXIV is the editor's printed number; the apparatus also records XXXVIII.

---

ambulationi laxamentum · room for walking

Ambulationi is dative: for the walk or passage. Laxamentum is the room provided. Vitruvius argues for use and appearance; his later rain example does not supply a measurable occupancy or loading limit.

Greek architectural vocabulary

The surrounding passage

τὸ πτερόν · πτεροῦ

ambulatio; pteroma

A wing becomes an architectural word for the space around a temple. Removing the inner colonnade joins the two narrow side lanes and the former shaft position into one broad strip. Dictionary headword.

The retained columns

ὁ κίων · κίονος

columna; exteriores ordines

Eight across each end and fifteen along each flank do not make 46 separate columns. The four corners belong to both descriptions. One circuit visits 42 distinct positions. Dictionary headword.

The omitted colonnade

δίπτερος → ψευδοδίπτερος

interiores ordines columnarum

Di- means two; pteron means wing. Pseudo- qualifies the double arrangement: its outer spacing remains, with the inner ring absent. The regular inner ring here has six by thirteen positions, or 34 distinct columns. Dictionary headword.

The enclosed temple room

ὁ ναός · ναοῦ

cella; parietes cellae

Naos can name the temple, and in architectural description its main room. This experiment simplifies the inner building to a block. The source restoration has porches and additional columns which this block does not represent. Dictionary headword.

Greek headwords explain the vocabulary; they are not a Greek original of the Latin. English explanations and translations are new.

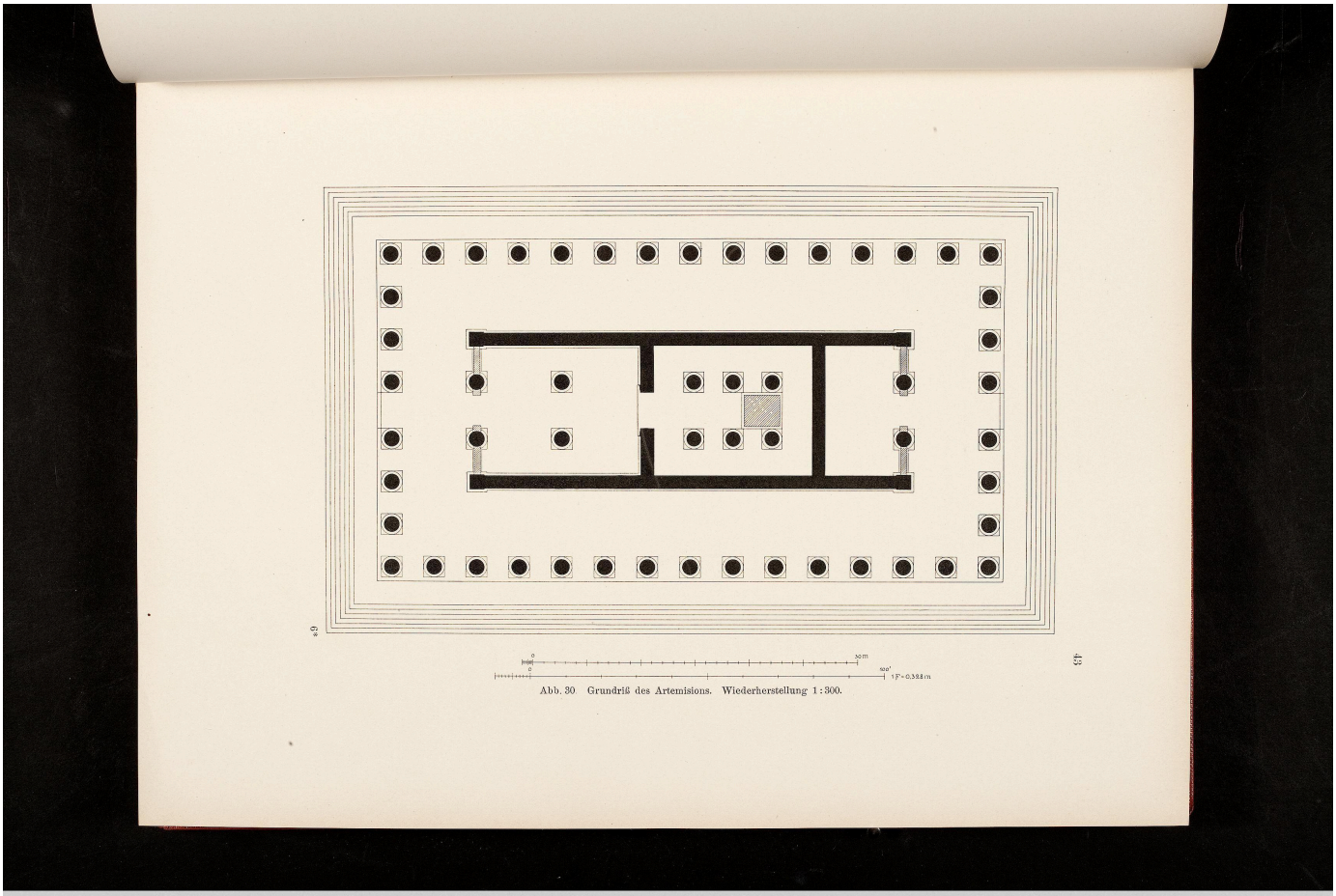
Scope and sources

The plans are a modern regular-grid experiment in lower shaft diameters (D), not a measured reconstruction of Magnesia. Equal gaps, a one-D wall and a simplified enclosed cella block are chosen. Porches, interior columns, bases, steps, roof supports and structural calculations are omitted. The 42 and 76 counts include only the model's perimeter rings. A clear strip is measured between shaft faces at column stations; actual bases would occupy more space. Geometry does not determine a textual reading, construction cost or crowd capacity.

Vitruvius III.2.6–7 and III.3.8, edited by F. Krohn (1912), pp. 63 and 66. Humann, Kohte and Watzinger, *Magnesia am Maeander* (1904), pp. 41, 43, 55: preserved-state plan, proposed restoration and capital photograph. Heidelberg University Library supplies the latter scans with the Public Domain Mark. The 1912 Latin scans are from Internet Archive. All page borders, legends, apparatus and library credits remain. Short excerpts join line-end hyphens and capitalize the numeral. New text, diagrams and code: MIT, Anton Soloviev, 2026. Each embedded image, font and license was verified by SHA-256 before this download was prepared.

A proposed restoration · figure 30, printed p. 43

Julius Kohte's restored plan in Humann, Kohte and Watzinger (1904). Wiederherstellung means restoration. Count eight across an end and fifteen along a flank. There are also porch and interior columns: the outer ring is not the whole building's column total. The centre opening at the ends is wider than neighbouring gaps. Original volume.

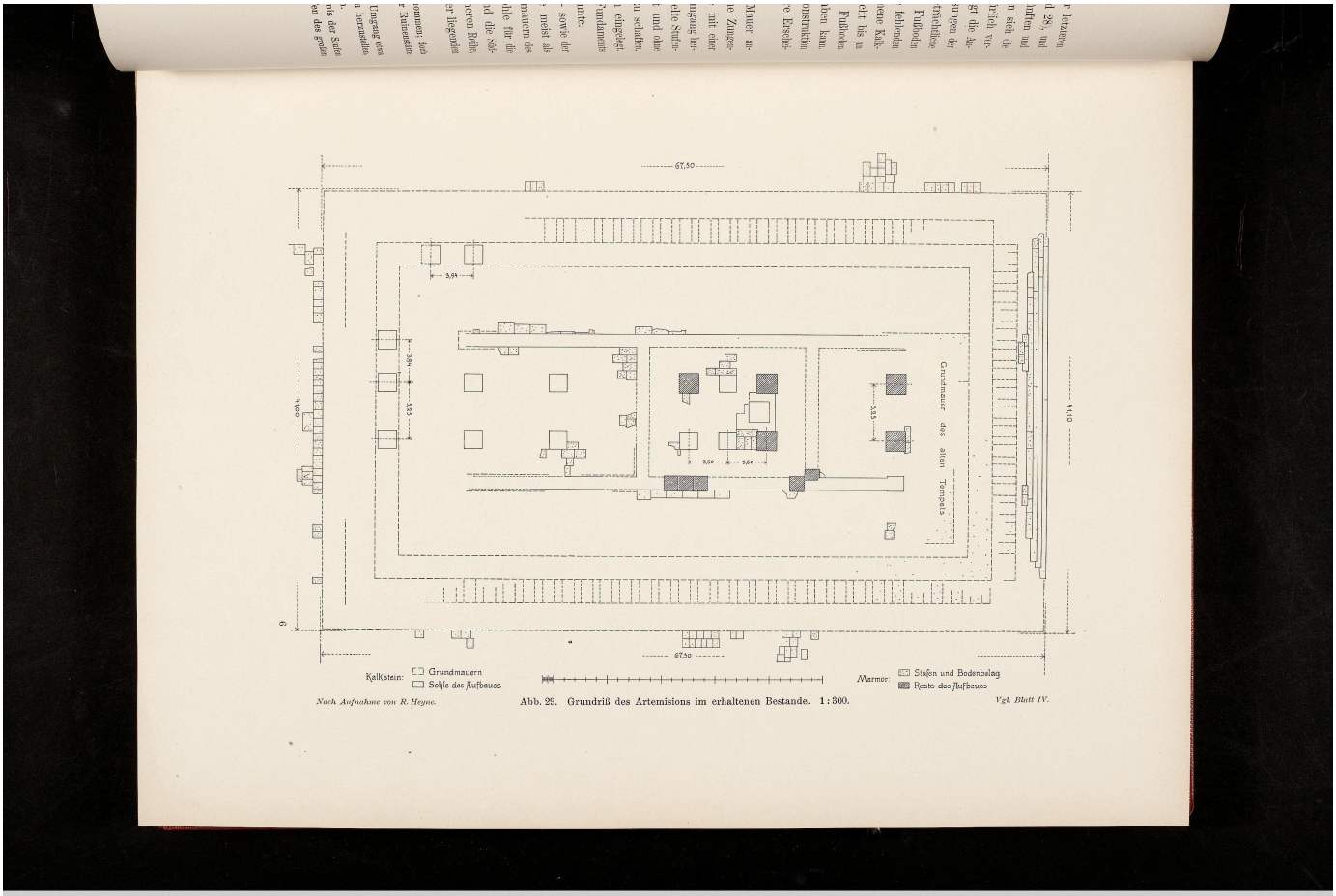


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The preserved state · figure 29, printed p. 41

The caption says im erhaltenen Bestande, in the preserved state, after R. Heyne's survey. Compare its different material symbols and discontinuous remains with the clean restoration. It records the excavation publication's assessment, not a current site survey or a complete standing temple. Original volume.

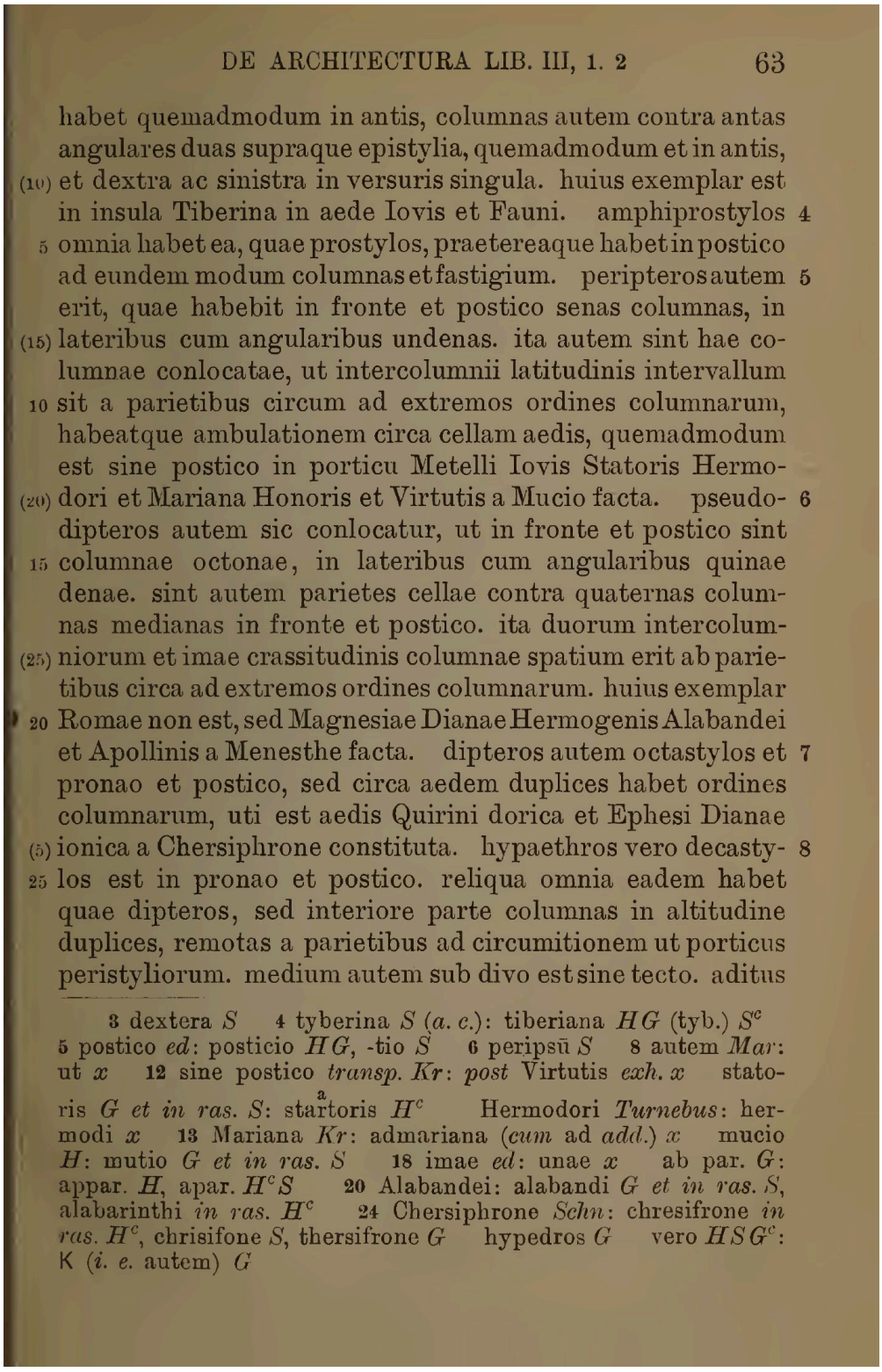


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The Latin prescription · Krohn, printed p. 63

Vitruvius III.2.6–7. The pseudodipteral prescription begins just above the middle of the page. The critical apparatus remains below. Our excerpts join line-end hyphens; English translations and grammar notes are new. Original volume.



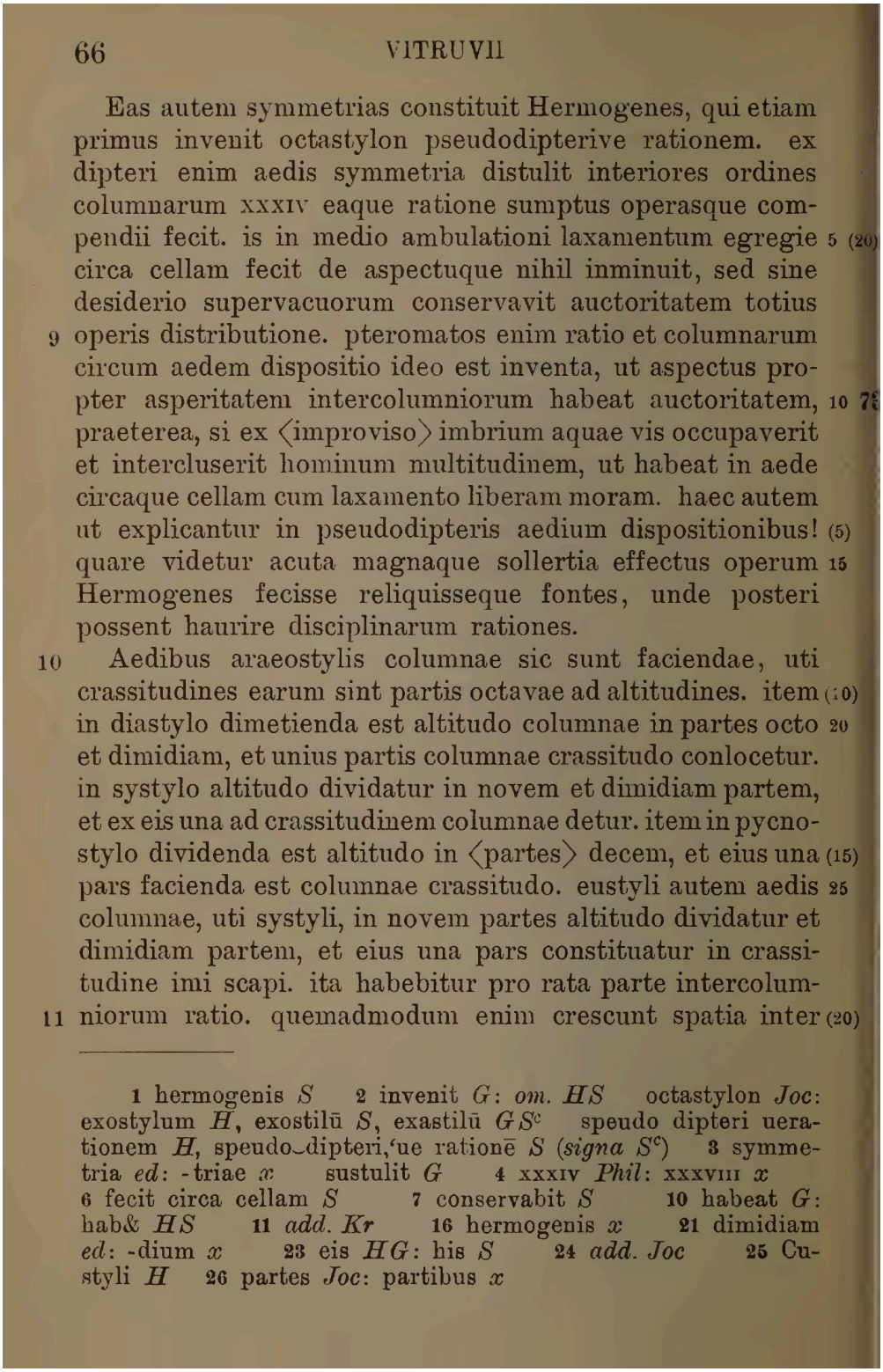
habet quemadmodum in antis, columnas autem contra antas angulares duas supraque epistylia, quemadmodum et in antis, et dextra ac sinistra in versuris singula. huius exemplar est in insula Tiberina in aede Iovis et Fauni. amphiprostylos 4 omnia habet ea, quae prostylos, praetereaque habet in postico ad eundem modum columnas et fastigium. peripteros autem 5 erit, quae habebit in fronte et postico senas columnas, in lateribus cum angularibus undenas. ita autem sint hae columnae conlocatae, ut intercolumnii latitudinis intervallum sit a parietibus circum ad extremos ordines columnarum, habeatque ambulationem circa cellam aedis, quemadmodum est sine postico in porticu Metelli Iovis Statoris Hermodori et Mariana Honoris et Virtutis a Mucio facta. pseudo- 6 dipteros autem sic conlocatur, ut in fronte et postico sint columnae octonae, in lateribus cum angularibus quinae denae. sint autem parietes cellae contra quaternas columnas medianas in fronte et postico. ita duorum intercolumniorum et imae crassitudinis columnae spatium erit ab parietibus circa ad extremos ordines columnarum. huius exemplar Romae non est, sed Magnesiae Dianae Hermogenis Alabandei et Apollinis a Menesthe facta. dipteros autem octastylos et 7 pronao et postico, sed circa aedem duplices habet ordines columnarum, uti est aedis Quirini dorica et Ephesi Dianae ionica a Chersiphrone constituta. hypaethros vero decasty- 8 los est in pronao et postico. reliqua omnia eadem habet quae dipteros, sed interiore parte columnas in altitudine duplices, remotas a parietibus ad circumitionem ut porticus peristyliorum. medium autem sub divo est sine tecto. aditus

3 dextera *S* 4 tyberina *S* (*a. c.*): tiberiana *HG* (tyb.) S^c 5 postico *ed*: posticio *HG*, -tio *S* 6 peripsū *S* 8 autem *Mar*: ut *x* 12 sine postico *transp. Kr*: *post* Virtutis *exh. x* statoris *G et in ras. S*: startoris H^c Hermodori *Turnebus*: hermodi *x* 13 Mariana *Kr*: admariana (*cum* ad *add.*) *x* mucio *H*: mutio *G et in ras. S* 18 imae *ed*: unae *x* ab par. *G*: appar. *H*, apar. H^cS 20 Alabandei: alabandi *G et in ras. S*, alabarinthi *in ras.* H^c 24 Chersiphrone *Schn*: chresifrone *in ras.* H^c, chrisifone *S*, thersifrone *G* hypedros *G* vero HSG^c: K (*i. e.* autem) *G*

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The disputed number · Krohn, printed p. 66

The main text prints xxxiv (34). In the apparatus, line 4 reads “xxxiv Phil: xxxviii x”. Krohn attributes 34 to Philander and records 38 in the textual tradition. A regular-ring calculation cannot decide whether that editorial reading should be accepted. Original volume.



66 VITRUVII

Eas autem symmetrias constituit Hermogenes, qui etiam primus invenit octastylon pseudodipterive rationem. ex dipteri enim aedis symmetria distulit interiores ordines columnarum XXXIV eaque ratione sumptus operasque compendii fecit. is in medio ambulationi laxamentum egregie (20) circa cellam fecit de aspectuque nihil inminuit, sed sine desiderio supervacuorum conservavit auctoritatem totius 9 operis distributione. pteromatos enim ratio et columnarum circum aedem dispositio ideo est inventa, ut aspectus propter asperitatem intercolumniorum habeat auctoritatem, praeterea, si ex ⟨improviso⟩ imbrium aquae vis occupaverit et intercluserit hominum multitudinem, ut habeat in aede circaque cellam cum laxamento liberam moram. haec autem ut explicantur in pseudodipteris aedium dispositionibus! (5) quare videtur acuta magnaque sollertia effectus operum Hermogenes fecisse reliquisseque fontes, unde posteri possent haurire disciplinarum rationes.

10 Aedibus araeostylis columnae sic sunt faciendae, uti crassitudines earum sint partis octavae ad altitudines. item (10) in diastylo dimetienda est altitudo columnae in partes octo et dimidiam, et unius partis columnae crassitudo conlocetur. in systylo altitudo dividatur in novem et dimidiam partem, et ex eis una ad crassitudinem columnae detur. item in pycnostylo dividenda est altitudo in ⟨partes⟩ decem, et eius una (15) pars facienda est columnae crassitudo. eustyli autem aedis columnae, uti systyli, in novem partes altitudo dividatur et dimidiam partem, et eius una pars constituatur in crassitudine imi scapi. ita habebitur pro rata parte intercolum-
11 niorum ratio. quemadmodum enim crescunt spatia inter (20)

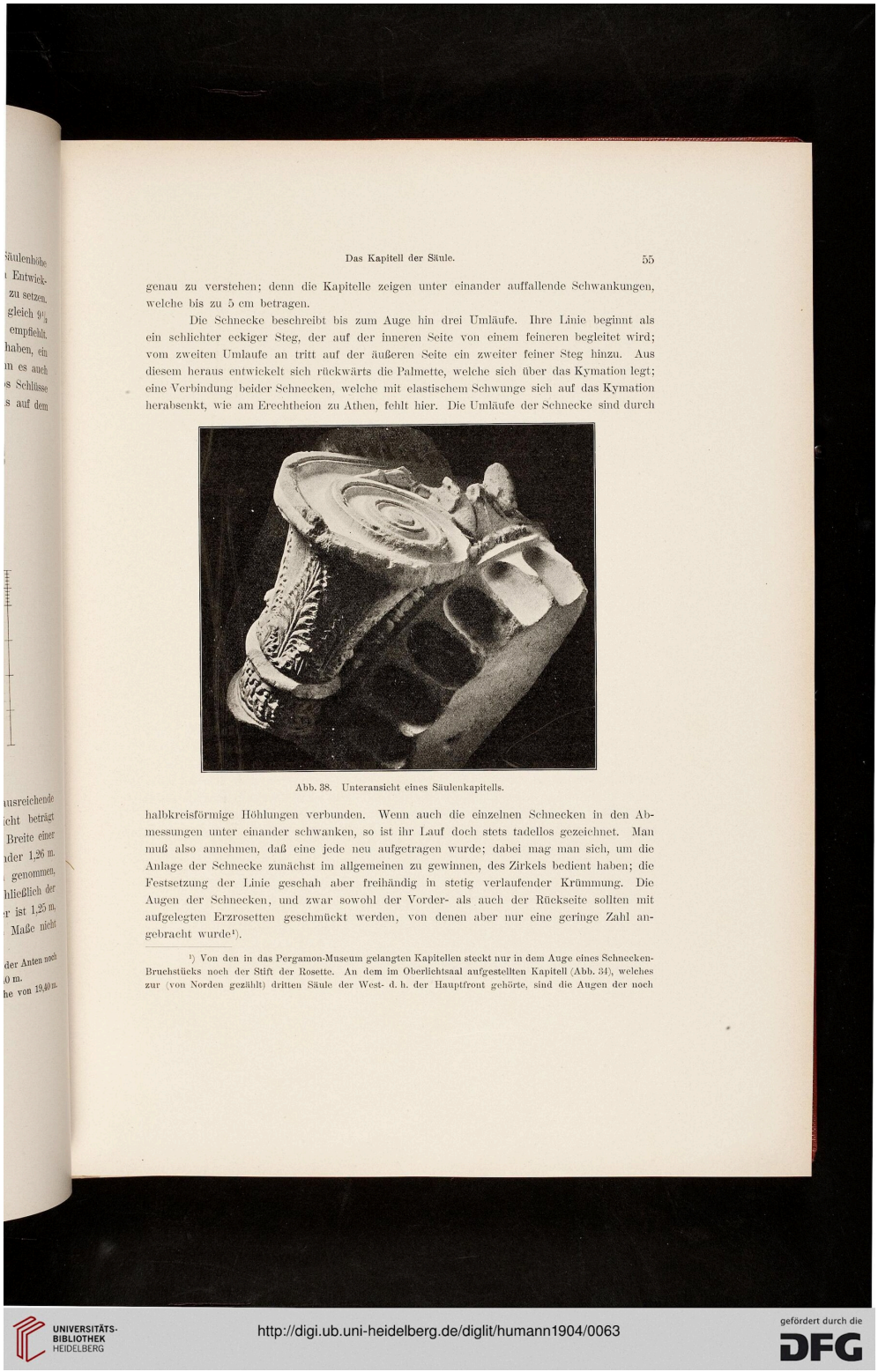
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1 hermogenis *S* 2 invenit *G*: *om. HS* octastylon *Joc*: exostylum *H*, exostilū *S*, exastilū GS^c speudo dipteri uerationem *H*, speudo⌣dipteri,ue rationē *S* (*signa* S^c) 3 symmetria *ed*: -triae *x* sustulit *G* 4 XXXIV *Phil*: XXXVIII *x* 6 fecit circa cellam *S* 7 conservabit *S* 10 habeat *G*: hab& *HS* 11 *add. Kr* 16 hermogenis *x* 21 dimidiam *ed*: -dium *x* 23 eis *HG*: his *S* 24 *add. Joc* 25 Custyli *H* 26 partes *Joc*: partibus *x*

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A surviving piece · figure 38, printed p. 55

The publication photographs the underside of an Ionic capital, with its volutes. It is an actual architectural fragment. Its photograph provides neither the missing inner colonnade nor proof that a dipteral predecessor once stood here. Original volume.



Das Kapitell der Säule. 55

genau zu verstehen; denn die Kapitelle zeigen unter einander auffallende Schwankungen, welche bis zu 5 cm betragen.

Die Schnecke beschreibt bis zum Auge hin drei Umläufe. Ihre Linie beginnt als ein schlichter eckiger Steg, der auf der inneren Seite von einem feineren begleitet wird; vom zweiten Umlaufe an tritt auf der äußeren Seite ein zweiter feiner Steg hinzu. Aus diesem heraus entwickelt sich rückwärts die Palmette, welche sich über das Kymation legt; eine Verbindung beider Schnecken, welche mit elastischem Schwunge sich auf das Kymation herabsenkt, wie am Erechtheion zu Athen, fehlt hier. Die Umläufe der Schnecke sind durch

Abb. 38. Unteransicht eines Säulenkapitells.

halbkreisförmige Höhlungen verbunden. Wenn auch die einzelnen Schnecken in den Abmessungen unter einander schwanken, so ist ihr Lauf doch stets tadellos gezeichnet. Man muß also annehmen, daß eine jede neu aufgetragen wurde; dabei mag man sich, um die Anlage der Schnecke zunächst im allgemeinen zu gewinnen, des Zirkels bedient haben; die Festsetzung der Linie geschah aber freihändig in stetig verlaufender Krümmung. Die Augen der Schnecken, und zwar sowohl der Vorder- als auch der Rückseite sollten mit aufgelegten Erzrosetten geschmückt werden, von denen aber nur eine geringe Zahl angebracht wurde[1]).

[1]) Von den in das Pergamon-Museum gelangten Kapitellen steckt nur in dem Auge eines Schnecken-Bruchstücks noch der Stift der Rosette. An dem im Oberlichtsaal aufgestellten Kapitell (Abb. 34), welches zur (von Norden gezählt) dritten Säule der West- d. h. der Hauptfront gehörte, sind die Augen der noch

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